

The Determinants of Credit Spread Changes¹

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Abstract

Using straight industrial bonds with quoted prices, we investigate the determinants of credit spread changes. We find the variables that should in theory determine credit spread changes in fact have limited explanatory power. Further, the residuals from this first-pass regression are highly cross-correlated, and principal components analysis strongly suggests they are driven by a single common factor. We investigate several macro-economic and financial variables as candidate proxies for this factor. We cannot, however, find any set of variables which explain this common systematic factor. Our results suggest the corporate bond market is a segmented market driven by corporate bond specific supply/demand shocks.

The relation between stock and bond returns has been widely studied at the aggregate level.¹ Recently, a few studies have investigated that relation at the individual firm level² and at the portfolio level.³ These studies focus on corporate bond returns, or yield changes. The main conclusions of these papers are: (1) high-grade bonds behave like Treasury bonds, and (2) low-grade bonds are more sensitive to stock returns.

The implications of these studies may be limited in many situations of interest, however. For example, hedge funds often take highly levered positions in corporate bonds while hedging away interest rate risk by shorting treasuries. As a consequence, their portfolios become extremely sensitive to changes in credit spreads rather than changes in bond yields. The distinction between changes in credit spreads and changes in corporate yields is significant: while an adjusted R^2 of 60% is obtained when regressing high-grade bond yield changes on Treasury yield changes and stock returns,⁴ we find that the R^2 falls to 5% when the dependent variable is credit spread changes. Hence, while much is known about yield changes, we have scant evidence about the determinants of credit spread changes.

Below, we investigate the determinants of credit spread changes. From a contingent-claims, or no-arbitrage standpoint, credit spreads obtain for two fundamental reasons: 1) there is a risk of default, and 2) in the event of default, the bondholder receives only a portion of the promised payments. Thus, we examine how changes in credit spreads respond to proxies for both changes in the probability of future default and for changes in the recovery rate.

Recent empirical studies have suggested that the corporate bond market tends to have relatively high transactions costs and low volume.⁵ These findings suggest looking beyond the pure contingent claims viewpoint for determinants of credit spread changes since one might

¹ See, for example, Campbell and Ammer (1993), Keim and Stambaugh (1986), Fama and French(1989) and Fama and French (1993).

² See, for example, Kwan (1996).

³ See, for example, Blume, Keim and Patel (1991) Cornell and Green (1991) .

⁴ See Kwan (1996).

⁵ At least in the period prior to 1997. See, for example, Schultz (1998), Hotchkiss and Ronen (1999), Chakravarty and Sarkar (1999).

expect to observe a liquidity premium. Thus, we also examine the ability of proxies for changes in liquidity to explain changes in credit spreads.

Our results are, in summary: although we consider a large number of variables that should measure changes in default probability, changes in recovery rate, or changes in liquidity, we can only explain about 25% of the observed credit spread changes. We find, however, that the residuals from this first-pass regression are highly cross-correlated, and principal components analysis strongly suggests they are driven by a single common factor. One important implication is that any omitted explanatory variables are likely not firm-specific. We therefore investigate several macroeconomic and financial variables as candidate proxies for this factor. We cannot, however, find any set of variables which explain this common systematic factor.

Our findings suggest that the corporate bond market is a segmented market driven by local supply/demand shocks. We note that a similar, but significantly smaller segmentation effect has been documented in the mortgage backed (Ginnie Mae) securities market by Boudoukh, Richardson, Stanton and Whitelaw (1997). Their 3-factor model explains over 90% of changes in Ginnie Mae yields, with the residual 10% mainly attributable to a single systematic factor that is segmented from the treasury market.⁶ In contrast, our multiple-factor model explains only about 25% of the variation in credit spreads, with most of the remainder attributable to a single systematic factor.

Existing literature on credit spread changes is extremely limited. Pedrosa and Roll (1998) document considerable comovement of credit spread changes among index portfolios of bonds from various industry, quality, and maturity groups. Note that this result by itself is not surprising, since theory predicts that all bonds *should* be affected by aggregate variables such as changes in the interest rate, changes in business climate, changes in market volatility, etc.

⁶ Their finding is unexpected since GNMA securities face no default risk but may be repaid early. If prepayment is rationally grounded in interest rates, then from a contingent-claims analysis, these bonds have prices and yields completely determined by the treasury market.

The particularly surprising part of our results is that, after controlling for these aggregate determinants, the systematic movement of credit spread changes still remains, and indeed, is the dominant factor.

The rest of the paper is organized as follows. In Section 1, we examine the theoretical determinants of credit spread changes. In Section 2, we discuss the data and define the proxies used. In Section 3, we analyze our results. In Section 4, we provide some evidence for the robustness of our results. We conclude in Section 5.

1 Theoretical Determinants of Credit Spread Changes

Models of risky debt basically fall into one of two categories: reduced-form models, or structural models of default. Reduced-form models specify the default process directly, and can most readily be interpreted as a jump process to default. This class of models has been investigated by Jarrow and Turnbull (1995), Jarrow, Lando and Turnbull (1997), Duffie and Singleton (1999), and others. Although this class of models has been very useful for fitting the observed credit spreads, its abstraction from the underlying firm value process makes it less useful for suggesting determinants of credit spread changes.

Structural models of default specify a particular firm value process, and assume that default is triggered when firm value falls to some explicit threshold. This default threshold is a function of the amount of debt outstanding. Introduced by Merton (1974) and further investigated by, among others, Black and Cox (1976), Leland (1994), and Longstaff and Schwartz (1995), this class of models determines both equity and debt prices in a self-consistent manner via arbitrage, or contingent-claims pricing. In these models, equity possesses characteristics similar to a call option, while the debt claim exhibits features analogous to a portfolio that has a claim to firm value, but that is short a call option. This interpretation is useful for predicting the determinants of credit spread changes.

To see this, consider a simple structural model of default. Firm value follows the risk-neutral process:

$$\frac{dV}{V} = (r - \delta) dt + \sigma dz^Q + \lambda (dq^Q - p dt) . \quad (1)$$

Here, V is firm value, r is the spot rate, δ is the firm payout rate, σ is firm volatility, λ is the size of a firm-value jump, and p is the risk-neutral probability, or intensity, of such a jump.

That is, the (risk-neutral) transition density of the jump process dq^Q is:

$$dq_t^Q = \begin{cases} 0 & \text{with probability } 1 - p dt \\ 1 & \text{with probability } p dt, \end{cases}$$

Default is assumed to occur the first time that firm value reaches a threshold K . Here, we implicitly assume that K is the amount of debt outstanding.

This structural framework suggests many factors that should affect changes in credit spreads:

1. Changes in the Spot Rate r .

As pointed out by Longstaff and Schwartz (1995), the static effect of a higher spot rate is to increase the risk-neutral drift of the firm value process. A higher drift reduces the incidence of default, and in turn, reduces the credit spreads. This prediction is borne out in their data. Further evidence is provided by Duffee (1998), who uses a sample restricted to non-callable bonds and finds a significant, albeit weaker, negative relationship between changes in credit spreads and interest rates.

2. Changes in Slope of Yield Curve.

Although the spot rate is the only interest-rate-sensitive factor that appears in (1), the spot rate process itself may depend upon other factors. For instance, Brennan and Schwartz (1979) introduce a model where short rate dynamics depends upon the long rate. Extending the logic of Longstaff and Schwartz (1995), if the short rate is expected

to mean-revert about the long rate, then an increase in the slope of the treasury curve should increase the expected future short rate, again leading to a decrease in credit spreads.

From a different perspective, a decrease in yield curve slope may imply a weakening economy. It is reasonable to believe that the expected recovery rate might decrease in times of recession.⁷ Once again, theory predicts that an increase in the treasury yield curve slope will create a decrease in credit spreads.

3. Changes in Leverage $\frac{K}{V}$

Within the structural framework, default is triggered when the leverage ratio approaches unity. Hence, it is clear that credit spreads are expected to increase with leverage. Likewise, credit spreads should decrease with the firm's equity return, all else equal.

4. Changes in Volatility σ

The contingent-claims approach implies that the debt claim has features similar to a short position in a call option. Option values increase with volatility. Hence, the model predicts that credit spreads should increase with volatility. This prediction is intuitive: increased volatility increases the probability of default.

5. Changes in the Probability or Magnitude of a Downward Jump $\{p, \lambda\}$

Implied volatility smiles in observed option prices suggest that markets do account for the probability of large negative jumps in firm value. Increases in either the probability or the magnitude of a negative jump should increase credit spreads.

6. Changes in the Business Climate

Even if the probability of default remains constant for a firm, changes in credit spreads

⁷ Fama and French (1989) find that credit spreads widen when economic conditions are weak.

can occur due to changes in the expected recovery rate. The expected recovery rate in turn should be a function of the overall business climate.⁸

2 Data

Our first objective is to investigate how well the variables that should in theory explain changes in credit spreads actually perform in practice. Here, we discuss the data used for estimating credit spreads and the explanatory variables.

1. Credit Spreads

The corporate bond data are obtained from Lehman Brothers via the Fixed Income (or Warga) Database. We use only quotes on non-callable, non-puttable debt of industrial firms; quotes are discarded whenever a bond has less than 4 years to maturity. Monthly observations are used for the period 7/88 – 12/97. Only observations with actual quotes are used, since it has been shown by Sarig and Warga (1989) that matrix prices are problematic.⁹ For each bond i and each month t , the credit spread CS_t^i is calculated relative to the contemporaneous Benchmark Treasury yield curve as provided by Datastream.

2. Treasury Rate level

We use Datastream’s monthly series of 10-year Benchmark Treasury rates r_t^{10} . To capture potential non-linearity we also include the squared level of the term structure $(r_t^{10})^2$.

3. Slope of Yield Curve

We define the slope of the yield curve as the difference between Datastream’s 10-year and 2-year Benchmark Treasury yields: $\text{slope}_t = r_t^{10} - r_t^2$. We interpret this proxy as

⁸ Altman and Kishore (1996) find that recovery rates are not constant over time.

⁹ Prices in the database are not all quotes— in months where no bid is posted, a matrix price is recorded instead as a “best guess”. Of 1209 bonds available with at least some concurrent stock return and leverage data, 688 have at least 24 actual monthly quotes and thus ultimately qualify for our sample.

both an indication of expectations of future short rates, and as an indication of overall economic health.

4. Firm Leverage

For each bond i , market values of firm equity from CRSP and book values of firm debt from Compustat are used to obtain leverage ratios lev_t^i , which we define as

$$\frac{\text{Book Value of Debt}}{\text{Market Value of Equity} + \text{Book Value of Debt}}.$$

Since debt levels are reported quarterly, linear interpolation is used to estimate monthly debt figures. We note that previous studies of yield changes have often used the firm's equity return to proxy for changes in the firm's health, rather than changes in leverage. For robustness, we also use each firm's monthly equity return ret_t^i , obtained from CRSP, as an explanatory variable.

5. Volatility

Ideally, changes in each firm's volatility might be extracted from changes in implied volatilities of its publicly traded options. Unfortunately, many of the firms we investigate lack publicly traded options. Instead, we use the best available substitute: changes in the VIX index (VIX_t), a weighted average of eight implied volatilities of near-the-money options on the OEX (S&P 100) index.¹⁰ These data are provided by the Chicago Board Options Exchange.

6. Jump Magnitudes and Probabilities

Changes in the probability and magnitude of a large negative jump in firm value should have a significant effect on credit spreads. This factor is rather difficult to proxy because

¹⁰The appropriate volatility input for structural models of default is typically that associated with the volatility of (debt + equity). We expect changes in the proposed proxy to be highly correlated with changes in this volatility.

historical occurrences of such jumps are rare enough to be of little value in predicting future probabilities and magnitude of such jumps. Therefore, we approach the problem with a forward-looking measure. In particular, we employ changes in the slope of the “smirk” of implied volatilities of options on S&P 500 futures to determine perceived changes in the probability of such jumps.

Options and futures prices were obtained from Bridge. Our proxy is constructed from at- and out-of-the money puts, and at- and in-the-money calls with the shortest maturity on the nearby S&P 500 futures contract. We first compute implied volatilities for each strike K using the standard Black and Scholes (1973) model. We then fit the linear-quadratic regression $\sigma(K) = a + bK + cK^2$, where K is the strike price. Our measure of slope is defined as: $\text{jump}_t = [\sigma(0.9F) - \sigma(F)]$, where F is the at-the money strike price, which equals the current futures price. We chose to look at the implied volatility at $K = .9F$ because we did not want to extrapolate the quadratic regression beyond the region where actual option prices were most typically observed.

Note that if there is a non-negligible probability of large negative jumps in firm value, then the appropriate hedging tool for corporate debt may not be the firm’s equity, but rather deep out-of-the-money puts on the firm’s equity. Assuming large negative jumps in firm value are highly correlated with market crashes, we hope to capture systematic changes in the market’s expectation of such events with this proxy. We expect that a steepening in the slope of the smirk will trigger an increase in credit spreads.

7. Changes in Business Climate

We use monthly S&P 500 returns r_t^{SP} as a proxy for the overall state of the economy. The data are obtained from CRSP.

Table 1: Explanatory variables and expected signs on the coefficients in the regression (2).

Variable	Description	Predicted Sign
Δr^{10}	Change in yield on 10-year Treasury	–
Δslope	Change in 10-year minus 2-year Treasury yields	–
ΔVIX	Change in implied volatility of S&P 500	+
Δlev^i	Change in firm leverage ratio	+
Δjump	Change in slope of Volatility Smile	–
r^{SP}	Return on S&P500	–

Table 1 summarizes the predicted sign of the correlation between changes in credit spreads and changes in the underlying variable.¹¹

3 The Empirical Test

3.1 Methodology

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression:

$$\begin{aligned} \Delta CS_t^i = & \alpha + \beta_1^i \Delta \text{lev}_t^i + \beta_2^i \Delta r_t^{10} + \beta_3^i (\Delta r_t^{10})^2 + \beta_4^i \Delta \text{slope}_t \\ & + \beta_5^i \Delta \text{VIX}_t + \beta_6^i r_t^{SP} + \beta_7^i \Delta \text{jump}_t + \epsilon_t^i. \end{aligned} \quad (2)$$

In Table 2, summary statistics of the distribution of coefficient estimates are presented. For ease of interpretation each bond is assigned to a leverage group based on the firm's average leverage ratio in months where the bond has quotes available. The leverage ratio groups ($<$

¹¹Notice these are predicted signs for univariate regressions. They may not necessarily hold in the multivariate regression we run below.

15%, 15-25%, 25-35%, 35-45%, 45-55%, >55%) are chosen to broadly replicate the top four quintiles and bottom two deciles of the sample. In Panels II and III, we show results for short- and long maturity subsamples. In the short subsample, quotes are discarded whenever a bond has more than 9 years to maturity, and in the long subsample, quotes are discarded whenever a bond has less than 12 years to maturity. Then, in each subsample and for each bond i still having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we re-estimate the regression (2).

Previous studies of corporate bonds have often used stock returns ret_t^i rather than changes in leverage to proxy for changes in the firm's health. Further, these studies have grouped bonds by rating rather than firm leverage. For robustness, we also investigate credit spread changes using this approach. We thus estimate the following regression:

$$\begin{aligned} \Delta CS_t^i = & \alpha + \beta_1^i \text{ret}_t^i + \beta_2^i \Delta r_t^{10} + \beta_3^i (\Delta r_t^{10})^2 + \beta_4^i \Delta \text{slope}_t \\ & + \beta_5^i \Delta \text{VIX}_t + \beta_6^i r_t^{SP} + \beta_7^i \Delta \text{jump}_t + \epsilon_t^i \end{aligned} \quad (3)$$

In Table 3, summary statistics of the distribution of coefficient estimates are presented. Each bond is assigned to a rating group based on the firm's average rating in months where the bond has quotes available. The bond rating is taken as the weaker of Moody's or S&P ratings whenever both are available. Maturity subsample results are also presented in Panels II and III of Table 3.

The results of models (2) and (3) are very similar. The adjusted R^2 ranges from 19%-25% for the whole sample, with a wider range of 17% - 34% when the sample is broken up into different maturities. The model performs worst when explaining variation in long-term, high-leverage bonds. This is a general feature of all our results. As we shall see, the model performs best when default risk is less important and most of the variation in credit spread is likely to be due to variations in the risk-free term structure.

3.2 Results

Most of the variables investigated in the regressions (2) and (3) have some ability to explain changes in credit spreads. Further, the signs of the estimated coefficients generally agree with theory. We summarize some of the major findings below.

1. From Tables 2 and 3 respectively, both the change in leverage Δlev_t^i and the firm equity return ret_t^i are statistically significant, with predicted sign, for most groups in the multivariate analyses. The economic significance, however, is rather weak. Indeed, the coefficient on the S&P 500 return is typically an order of magnitude more significant than that on the firm's own equity return. This is our first indication that monthly changes in firm-specific attributes are not the driving force in credit spread changes.

Sensitivity to changes in leverage also tends to increase as leverage does, but that result is more apparent in a univariate regression framework, shown in Tables 4 and 5. Tables 4 and 5 also demonstrate that the apparently weak explanatory power of firm-specific variables is not due to potential collinearity with the market return r_t^{SP} .

2. Consistent with the empirical findings of Longstaff and Schwartz (1995) and Duffee (1998), we find that an increase in the risk-free rate lowers the credit spread for all bonds. Furthermore, the sensitivity to interest rates increases monotonically across both leverage and rating groups. Once again, this finding can be explained by noting that an increase in drift decreases the probability of default, and the closer firms are to the default threshold, the more sensitive they are to this change.
3. Overall, convexity and slope of the term structure are not very significant either statistically or economically. Interestingly, in the short and long maturity subsamples, the coefficients on convexity and slope tend to be of opposite sign.
4. The change in VIX is statistically significant. As seen in panel II of Tables 2 and 3, it

appears to have its greatest economic impact for short maturity bonds' credit spreads. However, some of these results seem to be clouded by collinearity between S&P 500 returns and changes in the VIX index (sample correlation $-.52$).

To investigate further, we perform univariate regressions of credit spread changes on changes in the VIX, and find strong economic significance throughout. Looking more closely, Table 6 demonstrates that credit spreads respond asymmetrically to changes in implied volatility: increases in implied volatility dramatically impact credit spreads, whereas decreases do not.

While use of VIX in place of firm-specific volatility assumes a strong positive correlation between the two, this assumption does not seem to affect our results significantly. Indeed, our main result is that most of the credit spread innovation is unexplained, and that the residuals are highly correlated cross-sectionally. We note that if individual firm- and market-volatility changes are not highly correlated, then our proxy should bias our results away from finding residuals which are so systematic.

5. The return of the S&P 500 is extremely significant both economically and statistically. Estimated coefficients have about the same magnitude for all groups. As expected, it has a negative impact. A return of 1% for the S&P 500 is associated with a credit spread decrease of about 1.6 basis points.
6. The change in the steepness of the S&P 500 smirk, Δjump_t , is statistically and economically significant. The sign, as expected, indicates that an increase in the market's expected probability of a negative jump (as revealed by an increase in out-of-the-money put prices) triggers an increase in credit spreads. The latter behavior is relatively homogeneous across all bond groups.¹²

¹² Again, univariate regressions (not reported) suggest that some of the explanatory power of the change in smirk may also be captured by the S&P 500 return because of collinearity between the two

3.3 Further Analysis

Overall, the variables suggested by theory are significant both economically and statistically in explaining variations in individual firms' credit spreads. However, at most they capture only around 25% of the variation as measured by adjusted R^2 .

To better understand the nature of the remaining variation, we undertake an analysis of the residuals using principal components as follows. We assign each month's residuals to one of eighteen 'bins', determined by three maturity groups (< 12-year, 12- to 18-year, >18-year), and the six leverage groups. For each bin, we compute an average and then extract the principal components of the correlation matrix of the resulting eighteen time series. The results reveal that around 71% of the variation is due to the first component. Interestingly, this first component is basically an equally-weighted portfolio across quality and maturity groups. A much smaller (about 6%) second component is long in high-leverage debt and short low-leverage debt. Similar results obtain when the analysis is repeated using maturity and rating bins. Clearly, credit spreads contain a large systematic component not implied by structural model theories.

So far, we have only used factors suggested by traditional models of credit risk as regressors. If this list of factors were comprehensive, then the above results would suggest that corporate bond and equity markets are segmented to a large extent. That is, they would seem to be driven by different aggregate risk factors. If this conclusion holds, then using traditional models of credit risk to price and hedge risky debt is bound to be unsuccessful.

Of course, another possibility is that that we have omitted important systematic explanatory variables. In the following section, we investigate the robustness of our results by considering many other independent variables.

variables.

4 Robustness

To further investigate the idea that a significant portion of the corporate bond market is segmented from both equity and Treasury markets, we would like to show there is no obvious omitted systematic factor on the right-hand side of our regressions. While there can be no complete refutation of an omitted variables argument, we can strengthen our results by showing that they still obtain even after including many other variables with potential explanatory power on the right hand side of regressions like (2) and (3).

4.1 Methodology

To investigate the robustness of our results, we expand our regression model (2) to include additional explanatory variables. Further, we test for nonlinearities by introducing quadratic and cross terms into the regression. In addition to the seven previous variables, we include the following independent variables:

1. Measures of changes in liquidity

We construct two measures of changes in liquidity. First, we examine the relative frequency of quote vs. matrix prices in the Warga database. For each month t , we take the log change dfquote_t in the ratio of the number of quotes q_t to the total number of reported prices n_t , which includes matrix prices. We interpret a higher ratio of quotes as indicative of more liquidity, but the indicator is somewhat noisy as the overall scope of the database tends to increase over time. The second liquidity index is more general: a measurement of changes dtliq_t in on- minus off-the-run 30-year Treasury yields. If liquidity worsens and the gap between these two widens, this measure decreases.

2. Proxy for firm value process

For robustness we include both the individual firm's equity return ret_t^i and the change

in leverage Δlev_t^i as independent variables. Although they are highly correlated, it is conceivable that both together provide information that neither alone can.

3. Nonlinear effects

Above, we investigate first-order effects of changes in the spot rate on the change in credit spread. In fact, structural models of default predict that changes in credit spreads should be nonlinear functions of changes in leverage, interest rates, etc.¹³ We therefore include several nonlinear terms as regressors (squared and cubed change in leverage, S&P 500 return, change in risk-free rate, and cross-terms). Since most are insignificant, we report here the results only with the variables having significant contributing explanatory power: $(\Delta r^{10})^2$, $(\Delta r^{10})^3$.

4. SMB factor

Since the S&P 500 return proves relatively important for credit spreads, it makes sense to examine other equity return systematic factors, such as a Fama and French (1996) Small Minus Big (SMB) factor. We construct a similar size factor using the CRSP decile portfolios—our monthly smb_t is decile-one minus decile-ten.

5. Economic state variables

If there is mean-reverting behavior in spot rates, leverage, volatility, or credit spreads, then the beginning-of-month levels of those variables should contain information about the current month’s change in credit spreads. We thus include the time $t - 1$ levels of: spot rate r_{t-1}^{10} , firm leverage lev_{t-1}^i , VIX index VIX_{t-1} , and the default premium $defprem_{t-1}$. The latter is measured as monthly average BBB yield from all available quotes in the Warga database minus 10-year Treasury yield.

¹³ However, the structural models predict that these higher-order terms should be orders of magnitude smaller than the linear terms.

6. Leading effects of stocks on bonds

Since lagged values of equity return have been documented to have impact on changes in bond yields (e.g., Kwan (1996)), we include the 1-month lagged S&P 500 return r_{t-1}^{SP} as a regressor.

4.2 Results and Analysis

Incorporating the extra variables yields the following regression:

$$\begin{aligned} \Delta CS_t^i = & \alpha + \beta_1^i \Delta \text{lev}_t^i + \beta_2^i \Delta r_t^{10} + \beta_3^i (\Delta r_t^{10})^2 + \beta_4^i \Delta \text{slope}_t + \beta_5^i \Delta \text{VIX}_t + \beta_6^i r_t^{SP} + \beta_7^i \Delta \text{jump}_t \\ & + \beta_8^i \text{dfquote}_t + \beta_9^i \text{dtliq}_t + \beta_{10}^i \text{ret}_t^i + \beta_{11}^i (\Delta r_t^{10})^3 + \beta_{12}^i \text{smb}_t \\ & + \beta_{13}^i r_{t-1}^{10} + \beta_{14}^i \text{lev}_{t-1}^i + \beta_{15}^i \text{VIX}_{t-1} + \beta_{16}^i \text{defprem}_{t-1} + \beta_{17}^i r_{t-1}^{SP} + \epsilon_t^i \end{aligned} \quad (4)$$

As in the prior analyses, we estimate this regression on each individual corporate bond credit spread time series. We report in Tables 7 and 8 the results for the whole sample, and also subdivide it into both leverage and rating classes as before. The results for the subsamples by maturity are omitted for conciseness. The main finding is that, even though the added variables do contribute somewhat to our understanding of credit spread movements, they have not explained the systematic factor which was so prominent in the earlier residuals.

Although the adjusted R^2 from (4), at around 31%, is higher than in the first-pass regressions (2) and (3), a repetition of our principal component analysis shows that the residuals from regression (4) are still highly cross-correlated. The first principal component explains about 63% of the remaining variation, and the corresponding eigenvector is roughly equally weighted in all maturity and leverage (or ratings) groups.

Thus, the additional ten variables contribute little to explaining the systematic factor that drives the credit spread on bonds. Our major conclusion still holds: it appears that credit spread changes of individual bonds are mostly driven by an aggregate factor that is captured

neither in existing theoretical literature, nor by the “kitchen sink” exercise in regression (4).

Still, several of the second-pass regression results provide interesting insights about the determinants of credit spreads. We summarize these below:

1. Measures of liquidity changes

Both measures of liquidity changes ($dfquote_t$ and $dtliq_t$) have a negative sign, as expected. However, the difference between on- and off-the-run Treasury yields is both economically and statistically more significant. It indicates that a widening of 10 basis points in this difference is associated with an increase of about 4 basis points in credit spreads. This would be consistent with posited “flight to quality” effects.

2. Nonlinear effects

The cubic term in the change in interest rate is typically positive, but lacking in economic significance.

3. SMB factor

The coefficient on our SMB factor moves from negative to positive as leverage increases (rating decreases). The change in sign is not unexpected since firms tend to be quite large once having attained the uppermost credit ratings. However, the explanatory power is overall quite weak.

4. Economic state variables

The coefficient on default premium levels $defprem_{t-1}$ clearly reflects mean-reversion in credit spreads. The coefficient on the level of the risk-free rate r_{t-1}^{10} is negative and significant throughout, but this is a marginal effect. In a univariate context, reported in Table 9, the relation between changes in credit spreads and interest rate levels is uniformly positive, but there is no explanatory power. Finally, the coefficients on levels of leverage lev_{t-1}^i and VIX VIX_{t-1} have limited statistical significance.

5. Leading effects of stocks on bonds

The coefficients on lagged S&P 500 returns are negative and are statistically significant except for higher leverage (lower rated) bonds. In terms of economic significance, the effect is smaller, roughly 30% of the size of the current S&P 500 return.

4.3 Additional Evidence

To further check that our observation of a systematic factor is not spurious, we first repeat regression (4) with the addition of a single variable: $\Delta \text{defprem}_t$, the month t change in default premium (average BBB minus 10-year Treasury yield). In effect, we are then regressing the change in bond i 's credit spread on the change in credit spreads in general, and we expect to observe a very high R^2 indeed. To no surprise, the results (not reported) show adjusted R^2 of about 60% except for the lowest quality, non-investment grade groups.

Having included $\Delta \text{defprem}_t$ in the regression, we once again undertake principal components analysis of the residuals using the same methods as before. The results are telling. The first component now accounts for about 42% of the (now much smaller) remaining variation, and is no longer equally weighted across groups.

Overall, these tests reinforce the robustness of our results in the previous section about the existence of a bond market aggregate systematic factor that is independent of both equity markets and the risk-free term structure and that seems to drive most of the changes in credit spreads.

5 Conclusion

We investigate changes in credit spreads on individual bond yields. Our results show that the factors suggested by traditional models of default risk explain at most 25% of the variation in credit spreads as measured by the adjusted R^2 . Furthermore, an analysis of the residuals

shows that they are highly correlated and a principal component analysis reveals a single factor, equally weighted across all bonds, explains more than 70% of the remaining variation. In performing a kitchen sink regression in search of that additional factor, we are unable to provide any additional explanation in terms of major financial and economic variables.

Our conclusions fall into three main areas. First, in contrast to the predictions of structural models of default, aggregate factors appear much more important than firm-specific factors for credit spread changes. Second, our results suggest segmentation of the stock and bond markets. This has important implications for holders of corporate bond portfolios, and suggests the need for further research into the interaction between market and credit risks. Finally, a potential management implication of these results is that firm value processes may be affected by capital structure decisions; the Modigliani-Miller “constant risk class assumption” is inconsistent with our results.

It seems hard to reconcile our findings with the existing models of default risk, and, in particular, with the so-called structural models, based on contingent claims analysis initiated by Merton (1974). The latter predicts a relation between credit spreads and leverage, volatility and interest rates. Although early empirical tests of these models proved disappointing (Jones, Mason and Rosenfeld (1984), Kim, Ramaswamy and Sundaresan (1993)), recent extensions (Leland (1998), Mella-Barral and Perraudin (1997), Anderson and Sundaresan (1996)) have shown that introducing agency theory or dynamic capital structure decisions can help improve the fit of the *level* of the credit spread. However it seems unlikely that these extensions can generate the kind of correlation in *changes* in credit spread uncovered in our analysis.

A natural explanation for our findings is segmentation of bond and equity markets. Clearly if markets are segmented and different investors trade in bonds and stocks, then prices in those markets could be driven by independent demand/supply shocks in both markets. Notwithstanding, in that case one needs to explain why these markets are segmented, and if they are,

why equity and bonds do not react to the same aggregate factors.

Could imperfections in the bond market— e.g., transaction costs, liquidity— explain our findings? Recent studies by Schultz (1998), Chakravarty and Sarkar (1999), and Hotchkiss and Ronen (1999) conclude that the stock and bond markets are equally adept at efficiently incorporating new information into prices (i.e., “pricing efficiency”). At the same time, they also show that liquidity (as measured by trading volume and bid-ask spread) can have major effects on bond prices. So, potentially, an aggregate factor driving liquidity in the bond market could explain the common factor we are detecting. Our unsophisticated measures of liquidity, such as spread between on and off-the-run Treasuries, and frequency of quotes vs. matrix prices, may simply be inadequate at capturing this factor.

Our findings appear to highlight a shortcoming of existing theoretical models of default risk. Except for interest rates, structural models of default focus on firm-specific factors driving the default event. Hence, they uniformly predict that the explanatory power of firm-specific measures (e.g., equity return, firm leverage) should swamp that of aggregate measures (e.g., market return).¹⁴ Empirically, we infer that most of the variation in credit spreads of individual bonds is explained by an aggregate factor common to all bonds. Thus, our paper suggests the need for further work on the interaction between market risk and credit risk— i.e., general equilibrium models embedding default risk.¹⁵

¹⁴ Indeed, we have justified including the S&P 500 return in our regressions as a proxy for changes in expected recovery rates, even though there is limited empirical support for such a claim.

¹⁵ See Chang and Sundaresan (1999) for first attempts in this direction.

Table 2: Structural Model Determinants of Credit Spread Changes by Leverage Group

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression: $\Delta CS_t^i = \alpha + \beta_1^i \Delta \text{lev}_t^i + \beta_2^i \Delta r_t^{10} + \beta_3^i (\Delta r_t^{10})^2 + \beta_4^i \Delta \text{slope}_t + \beta_5^i \Delta \text{VIX}_t + \beta_6^i r_t^{SP} + \beta_7^i \Delta \text{jump}_t + \epsilon_t^i$. Quotes are discarded whenever a bond has less than 4 years to maturity. Average OLS parameter estimates are reported in Panel I. Panel II shows averages for a short maturity subsample where quotes are discarded whenever a bond has more than 9 years to maturity. Panel III shows averages for a long maturity subsample where quotes are discarded whenever a bond has less than 12 years to maturity. Associated t-statistics for each average appear immediately beneath.

	Leverage Groups					
	<15%	15-25%	25-35%	35-45%	45-55%	>55%
I. Whole Sample						
intercept	.022	.016	.013	.013	.010	-.002
t	8.76	10.00	6.57	4.59	2.73	-0.20
Δlev_t^i	-.005	.007	.003	.004	.008	.033
	-1.74	4.89	1.86	2.02	3.35	3.75
Δr_t^{10}	-.124	-.140	-.181	-.215	-.215	-.342
	-17.84	-30.23	-18.93	-17.63	-11.93	-6.15
$(\Delta r_t^{10})^2$	-.010	-.001	.009	.048	.004	.164
	-0.54	-0.05	0.67	2.40	0.10	2.31
Δslope_t	.006	.001	-.028	.008	.004	-.033
	0.30	0.07	-2.29	0.48	0.15	-0.73
ΔVIX_t	.001	.002	.003	-.001	.005	.001
	0.82	3.44	2.85	-0.94	2.65	0.11
r_t^{SP}	-.016	-.015	-.016	-.017	-.016	-.019
	-21.00	-29.56	-22.68	-15.60	-10.65	-6.85
Δjump_t	.004	.004	.003	.002	.004	.003
	16.86	18.50	7.76	5.83	7.87	1.88
adjusted R^2	0.244	0.23	0.211	0.216	0.197	0.192
N	100	162	138	123	91	74
II. Short Maturity Subsample						
intercept	.023	.019	.009	.015	.006	-.008
	10.02	9.64	2.93	3.41	1.17	-0.58
Δlev_t^i	-.003	.009	.004	.003	.002	.042
	-0.77	5.00	1.51	1.14	0.76	3.04
Δr_t^{10}	-.141	-.138	-.202	-.226	-.235	-.414
	-20.65	-19.97	-11.68	-12.10	-7.68	-4.78
$(\Delta r_t^{10})^2$	-.046	-.032	-.020	.012	-.046	.165
	-2.65	-1.97	-0.89	0.37	-0.98	1.42
Δslope_t	.043	.031	-.045	.020	.031	.005
	2.15	2.87	-1.63	0.67	0.88	0.07
ΔVIX_t	.004	.004	.005	.001	.009	.002
	2.60	3.40	3.39	0.37	3.20	0.26
r_t^{SP}	-.017	-.015	-.018	-.018	-.019	-.020
	-24.03	-22.04	-14.43	-11.25	-10.53	-4.90
Δjump_t	.005	.005	.004	.002	.005	.004
	13.52	15.04	4.70	3.15	4.91	1.63
adjusted R^2	.317	.284	.264	.248	.199	.197
N	53	91	65	64	47	46
III. Long Maturity Subsample						
intercept	.010	.013	.006	.014	.007	.005
	1.89	3.98	3.54	4.25	1.24	1.48
Δlev_t^i	-.008	.004	.004	.002	.015	.013
	-1.68	1.39	1.90	0.78	3.32	6.22
Δr_t^{10}	-.095	-.161	-.156	-.200	-.210	-.211
	-5.86	-18.16	-12.75	-10.34	-9.93	-8.01
$(\Delta r_t^{10})^2$	.076	.057	.056	.055	.091	.143
	1.67	2.43	3.93	2.20	1.82	5.15
Δslope_t	-.029	-.028	-.035	-.019	.003	-.088
	-0.68	-2.45	-2.68	-0.89	0.07	-3.58
ΔVIX_t	-.002	.001	.003	-.001	.002	-.002
	-1.35	0.40	1.90	-0.78	0.51	-1.49
r_t^{SP}	-.014	-.015	-.012	-.017	-.013	-.017
	-14.70	-14.00	-9.87	-11.13	-4.72	-7.98
Δjump_t	.004	.004	.003	.003	.004	.002
	9.22	10.63	6.26	4.87	7.15	3.30
adjusted R^2	.205	.213	.196	.201	.216	.191
N	33	54	50	45	33	27

Table 3: Structural Model Determinants of Credit Spread Changes by Rating Group

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression: $\Delta CS_t^i = \alpha + \beta_1^i \text{ret}_t^i + \beta_2^i \Delta r_t^{10} + \beta_3^i (\Delta r_t^{10})^2 + \beta_4^i \Delta \text{slope}_t + \beta_5^i \Delta \text{VIX}_t + \beta_6^i r_t^{SP} + \beta_7^i \Delta \text{jump}_t + \epsilon_t^i$. Quotes are discarded whenever a bond has less than 4 years to maturity. Average OLS parameter estimates are reported in Panel I. Panel II shows averages for a short maturity subsample where quotes are discarded whenever a bond has more than 9 years to maturity. Panel III shows averages for a long maturity subsample where quotes are discarded whenever a bond has less than 12 years to maturity. Associated t-statistics for each average appear immediately beneath.

	Rating Groups					
	AAA	AA	A	BBB	BB	B
I. Whole Sample						
intercept	.021	.016	.011	.018	.009	-.033
t	2.89	8.17	10.78	9.44	1.82	-0.67
ret_t^i	.002	.000	-.001	-.002	-.003	-.018
t	2.11	0.15	-2.67	-4.15	-4.58	-2.75
Δr_t^{10}	-.109	-.150	-.151	-.159	-.296	-.862
t	-7.15	-17.99	-27.73	-26.03	-14.74	-4.36
$(\Delta r_t^{10})^2$	-.039	-.012	.037	-.014	.095	.568
t	-0.52	-0.76	3.94	-1.02	2.15	1.19
Δslope_t	.042	.009	-.017	.027	-.060	.048
t	0.55	0.70	-1.90	2.83	-1.92	0.36
ΔVIX_t	.002	.004	.002	.002	.000	-.029
t	0.62	2.92	4.44	2.88	-0.11	-0.79
r_t^{SP}	-.016	-.015	-.014	-.014	-.023	-.043
t	-14.36	-18.50	-37.00	-21.22	-9.82	-3.65
Δjump_t	.003	.004	.003	.003	.004	.005
t	2.83	10.24	13.57	12.98	6.62	0.98
adjusted R^2	.222	.293	.234	.194	.197	.275
N	4	56	275	245	90	18
II. Short Maturity Subsample						
intercept	.031	.018	.014	.016	.007	-.041
t	5.02	5.74	8.33	5.82	0.94	-0.70
ret_t^i	.000	.000	-.001	-.001	-.003	-.019
t	-0.24	0.47	-2.72	-2.28	-2.70	-2.51
Δr_t^{10}	-.111	-.156	-.163	-.150	-.322	-.909
t	-5.60	-14.39	-18.98	-14.76	-10.73	-3.86
$(\Delta r_t^{10})^2$	-.123	-.060	-.015	-.031	.040	.607
t	-1.10	-2.65	-1.19	-1.89	0.65	1.05
Δslope_t	.168	.028	.001	.052	-.032	.072
t	2.16	1.34	0.10	3.45	-0.67	0.44
ΔVIX_t	.006	.005	.006	.006	.001	-.038
t	0.82	2.63	6.50	4.49	0.35	-0.87
r_t^{SP}	-.015	-.016	-.015	-.015	-.026	-.044
t	-7.75	-18.37	-22.56	-18.76	-7.62	-3.31
Δjump_t	.002	.004	.003	.004	.005	.009
t	0.97	6.99	8.46	8.85	4.60	1.51
adjusted R^2	.232	.341	.277	.235	.200	.301
N	2	34	139	120	56	15
III. Long Maturity Subsample						
intercept	.009	.014	.007	.015	.008	-.031
t	8.66	4.23	3.71	5.07	1.60	-2.61
ret_t^i	.004	-.001	.000	-.003	-.004	-.001
t	9.38	-0.89	-1.25	-3.53	-3.65	-0.19
Δr_t^{10}	-.096	-.159	-.143	-.178	-.234	-.611
t	-14.97	-10.33	-16.11	-18.05	-10.09	-5.61
$(\Delta r_t^{10})^2$	.074	.020	.078	.049	.176	.270
t	2.66	0.87	4.35	2.63	3.48	2.06
Δslope_t	-.074	-.003	-.039	.000	-.083	-.197
t	-3.24	-0.20	-2.72	0.02	-2.78	-0.88
ΔVIX_t	-.001	.003	.001	-.001	.000	.007
t	-0.63	1.76	0.77	-1.14	0.02	0.83
r_t^{SP}	-.016	-.013	-.012	-.014	-.020	-.027
t	-20.50	-6.57	-21.93	-13.22	-5.43	-2.49
Δjump_t	.004	.004	.003	.003	.004	-.003
t	230.43	5.39	10.73	9.95	4.23	-1.71
adjusted R^2	.179	.265	.224	.180	.165	.302
N	2	16	114	79	28	3

Table 4: Relation Between Changes in Credit Spreads and Changes in Leverage

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression: $\Delta CS_t^i = \alpha + \beta_1^i \Delta \text{lev}_t^i + \epsilon_t^i$. Quotes are discarded whenever a bond has less than 4 years to maturity. Average OLS parameter estimates are reported in Panel I. Panel II shows averages for a short maturity subsample where quotes are discarded whenever a bond has more than 9 years to maturity. Panel III shows averages for a long maturity subsample where quotes are discarded whenever a bond has less than 12 years to maturity. Associated t-statistics for each average appear immediately beneath.

		Leverage Groups					
		<15%	15-25%	25-35%	35-45%	45-55%	>55%
I. Whole Sample							
intercept		.001	.000	-.003	-.004	-.005	.005
t		1.21	-0.01	-3.38	-2.54	-2.46	1.36
Δlev_t^i		.012	.015	.010	.011	.016	.035
		3.87	10.30	7.07	5.38	7.17	5.21
adjusted R^2		.003	.028	.011	.032	.051	.065
N		100	162	138	123	91	74
II. Short Maturity Subsample							
intercept		-.004	-.002	-.008	-.007	-.015	.006
		-3.24	-1.98	-5.40	-2.86	-4.40	1.03
Δlev_t^i		.016	.016	.014	.011	.013	.042
		3.45	10.03	5.19	5.27	5.55	4.25
adjusted R^2		.001	.025	.024	.033	.030	.072
N		53	91	65	64	47	46
III. Long Maturity Subsample							
intercept		.001	.000	-.001	.000	.003	.000
		1.06	-0.15	-1.10	-0.24	0.95	-0.12
Δlev_t^i		.006	.012	.007	.007	.021	.018
		1.60	4.10	4.47	2.47	4.20	7.66
adjusted R^2		-.008	.016	.005	.021	.084	.055
N		33	54	50	45	33	27

Table 5: Relation Between Changes in Credit Spreads and Firm Equity Returns

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression: $\Delta CS_t^i = \alpha + \beta_1^i \text{ret}_t^i + \epsilon_t^i$. Quotes are discarded whenever a bond has less than 4 years to maturity. Average OLS parameter estimates are reported in Panel I. Panel II shows averages for a short maturity subsample where quotes are discarded whenever a bond has more than 9 years to maturity. Panel III shows averages for a long maturity subsample where quotes are discarded whenever a bond has less than 12 years to maturity. Associated t-statistics for each average appear immediately beneath.

	Rating Groups					
	AAA	AA	A	BBB	BB	B
I. Whole Sample						
intercept	.007	.003	.003	.001	-.007	.022
t	3.67	4.76	6.06	1.10	-2.69	1.41
ret_t^i	-.003	-.003	-.003	-.004	-.005	-.014
	-1.97	-7.78	-14.70	-9.22	-7.39	-3.82
adjusted R^2	.004	.018	.030	.040	.047	.115
N	4	56	275	245	90	18
II. Short Maturity Subsample						
intercept	.009	.002	.001	-.004	-.015	.020
	2.34	2.04	1.58	-2.73	-3.76	1.10
ret_t^i	-.005	-.003	-.003	-.003	-.005	-.015
	-2.62	-5.92	-12.97	-8.82	-4.86	-3.86
adjusted R^2	.027	.019	.033	.035	.033	.116
N	2	34	139	120	56	15
III. Long Maturity Subsample						
intercept	.004	.003	.002	.003	.000	-.011
	21.06	3.88	2.52	2.23	0.12	-0.97
ret_t^i	-.001	-.002	-.002	-.004	-.005	-.001
	-3.96	-5.36	-8.03	-5.21	-5.66	-0.18
adjusted R^2	-.016	.004	.011	.050	.067	.079
N	2	16	114	79	28	3

Table 6: Relation Between Changes in Credit Spreads and Changes in VIX by Leverage Group

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression: $\Delta CS_t^i = \alpha + \beta_1^i \Delta VIX_t d_t + \beta_2^i \Delta VIX_t (1 - d_t) + \epsilon_t^i$, where $d_t = 1$ if $\Delta VIX_t > 0$, and 0 otherwise. Quotes are discarded whenever a bond has less than 4 years to maturity. Average OLS parameter estimates are reported in Panel I. Panel II shows averages for a short maturity subsample where quotes are discarded whenever a bond has more than 9 years to maturity. Panel III shows averages for a long maturity subsample where quotes are discarded whenever a bond has less than 12 years to maturity. Associated t-statistics for each average appear immediately beneath.

	Leverage Groups					
	<15%	15-25%	25-35%	35-45%	45-55%	>55%
I. Whole Sample						
intercept	-.015	-.019	-.017	-.020	-.021	-.021
t	-8.76	-8.30	-7.40	-5.75	-4.46	-3.36
positive ΔVIX_t	.014	.016	.014	.013	.016	.026
	20.27	14.58	11.54	8.49	7.72	7.55
negative ΔVIX_t	.001	.001	.003	.001	.005	.005
	1.15	0.32	2.15	0.27	2.09	1.34
adjusted R^2	.041	.048	.029	.023	.029	.030
N	100	162	138	123	91	74
II. Short Maturity Subsample						
intercept	-.021	-.022	-.027	-.033	-.039	-.022
t	-7.99	-9.19	-5.76	-6.40	-4.71	-2.26
positive ΔVIX_t	.018	.018	.019	.019	.024	.031
	14.50	17.89	7.27	8.70	7.65	5.89
negative ΔVIX_t	.004	.004	.003	-.001	.005	.010
	1.73	2.12	1.46	-0.22	1.28	1.74
adjusted R^2	.075	.060	.046	.045	.054	.043
N	53	91	65	64	47	46
III. Long Maturity Subsample						
intercept	-.016	-.022	-.007	-.008	-.004	-.023
t	-5.64	-4.10	-3.90	-1.51	-0.62	-4.69
positive ΔVIX_t	.011	.014	.009	.008	.008	.013
	11.69	5.50	6.84	4.00	2.68	4.58
negative ΔVIX_t	-.002	-.005	.004	.003	.007	.000
	-1.47	-1.30	3.48	0.96	2.23	-0.18
adjusted R^2	.017	.041	.011	.015	.009	.013
N	33	54	50	45	33	27

Table 7: Additional Determinants of Credit Spread Changes by Leverage Group

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression: $\Delta CS_t^i = \alpha + \beta_1^i \Delta \text{lev}_t^i + \beta_2^i \Delta r_t^{10} + \beta_3^i (\Delta r_t^{10})^2 + \beta_4^i \Delta \text{slope}_t + \beta_5^i \Delta \text{VIX}_t + \beta_6^i r_t^{SP} + \beta_7^i \Delta \text{jump}_t + \beta_8^i \text{dfquote}_t + \beta_9^i \Delta \text{dtliq}_t + \beta_{10}^i \text{ret}_t^i + \beta_{11}^i (\Delta r_t^{10})^3 + \beta_{12}^i \text{smb}_t + \beta_{13}^i r_{t-1}^{10} + \beta_{14}^i \text{lev}_{t-1}^i + \beta_{15}^i \text{VIX}_{t-1} + \beta_{16}^i \text{defprem}_{t-1} + \beta_{17}^i r_{t-1}^{SP} + \epsilon_t^i$. Quotes are discarded whenever a bond has less than 4 years to maturity. Average OLS parameter estimates are reported in Panel I. Associated t-statistics for each average appear immediately beneath.

	Leverage Groups					
	<15%	15-25%	25-35%	35-45%	45-55%	>55%
I. Whole Sample						
intercept	.795	.565	.624	.523	.061	-.294
t	10.09	8.69	2.89	6.35	0.46	-0.98
Δlev_t^i	-.002	.016	-.002	.011	-.009	.002
	-0.29	3.57	-0.15	2.25	-0.80	0.15
Δr_t^{10}	-.216	-.166	-.197	-.301	-.329	-.492
	-10.62	-11.43	-8.73	-12.54	-9.37	-4.34
$(\Delta r_t^{10})^2$	-.121	-.045	-.041	-.027	.003	-.161
	-3.83	-1.95	-1.29	-0.74	0.06	-1.71
Δslope_t	.044	.062	.033	.077	.053	-.023
	2.14	6.08	1.92	4.24	1.83	-0.25
ΔVIX_t	.002	.006	.006	.005	.020	.027
	1.43	3.47	3.29	1.91	5.45	4.61
r_t^{SP}	-.014	-.014	-.014	-.013	-.009	-.014
	-14.37	-16.00	-12.61	-7.62	-4.68	-2.89
Δjump_t	.004	.004	.003	.002	.005	.004
	9.80	11.83	5.74	3.96	6.15	1.64
dfquote_t	-1.032	-.093	-.420	-.498	-.007	-.480
	-2.37	-0.36	-0.98	-1.18	-0.01	-0.48
Δdtliq_t	-.376	-.223	-.195	-.345	-.425	-.201
	-5.35	-3.92	-2.34	-3.94	-3.80	-1.25
ret_t^i	.061	.178	.002	.143	-.558	-.663
	1.00	2.40	0.02	1.13	-1.93	-2.59
$(\Delta r_t^{10})^3$	.662	.038	.193	.575	.417	.426
	3.78	0.39	1.20	4.15	2.71	1.47
smb_t	.120	-.006	.032	-.018	-.240	-.649
	3.20	-0.18	0.59	-0.40	-2.38	-2.09
r_{t-1}^{10}	-.051	-.045	-.046	-.054	-.057	-.086
	-7.53	-9.14	-5.32	-7.85	-5.46	-3.13
lev_{t-1}^i	.320	.258	.077	.426	.594	.571
	1.74	3.03	0.30	3.88	4.95	1.01
VIX_{t-1}	.001	.006	.005	.009	.025	.046
	0.68	2.35	1.73	2.71	4.68	3.55
defprem_{t-1}	-.459	-.353	-.350	-.431	-.295	-.234
	-13.59	-14.88	-6.32	-13.49	-6.22	-1.44
r_{t-1}^{SP}	-.005	-.005	-.005	-.005	-.003	-.001
	-6.10	-6.72	-5.33	-4.23	-1.77	-0.12
adjusted R^2	0.401	0.331	0.317	0.324	0.281	0.295
N	100	162	138	123	91	74

Table 8: Additional Determinants of Credit Spread Changes by Rating Group

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression: $\Delta CS_t^i = \alpha + \beta_1^i \Delta \text{lev}_t^i + \beta_2^i \Delta r_t^{10} + \beta_3^i (\Delta r_t^{10})^2 + \beta_4^i \Delta \text{slope}_t + \beta_5^i \Delta \text{VIX}_t + \beta_6^i r_t^{SP} + \beta_7^i \Delta \text{jump}_t + \beta_8^i \text{dfquote}_t + \beta_9^i \Delta \text{dtliq}_t + \beta_{10}^i \text{ret}_t^i + \beta_{11}^i (\Delta r_t^{10})^3 + \beta_{12}^i \text{smb}_t + \beta_{13}^i r_{t-1}^{10} + \beta_{14}^i \text{lev}_{t-1}^i + \beta_{15}^i \text{VIX}_{t-1} + \beta_{16}^i \text{defprem}_{t-1} + \beta_{17}^i r_{t-1}^{SP} + \epsilon_t^i$. Quotes are discarded whenever a bond has less than 4 years to maturity. Average OLS parameter estimates are reported in Panel I. Associated t-statistics for each average appear immediately beneath.

	Rating Groups					
	AAA	AA	A	BBB	BB	B
I. Whole Sample						
intercept	.317	1.030	.467	.505	.075	-.695
t	0.61	2.01	9.63	6.88	0.55	-0.65
Δlev_t^i	.023	-.018	.006	.008	.006	-.023
t	1.22	-0.72	1.81	2.11	0.77	-0.44
Δr_t^{10}	-.083	-.136	-.172	-.254	-.434	-1.246
t	-1.91	-4.68	-16.79	-16.72	-10.23	-3.02
$(\Delta r_t^{10})^2$	-.105	-.090	-.018	-.076	-.014	-.535
t	-3.18	-4.21	-1.13	-3.29	-0.19	-1.58
Δslope_t	.051	.095	.032	.061	.026	.008
t	1.26	3.63	2.93	4.31	0.75	0.02
ΔVIX_t	.002	.005	.004	.008	.021	.067
t	0.38	2.00	5.22	5.09	3.92	4.34
r_t^{SP}	-.017	-.012	-.014	-.011	-.015	-.019
t	-8.85	-13.11	-22.54	-14.05	-5.05	-1.00
Δjump_t	.005	.003	.003	.004	.004	.007
t	4.60	4.12	10.96	8.51	5.15	0.82
dfquote_t	2.363	-.513	-.084	-.381	-.608	-4.623
t	2.28	-1.15	-0.49	-1.37	-0.67	-1.26
Δdtliq_t	-.362	-.061	-.261	-.335	-.451	.135
t	-3.02	-0.50	-5.38	-5.14	-3.71	0.48
ret_t^i	.268	-.257	.084	-.021	-.235	-1.692
t	2.78	-0.95	1.10	-0.26	-1.07	-1.50
$(\Delta r_t^{10})^3$	-.329	-.060	.154	.519	.575	1.246
t	-1.60	-0.27	2.20	4.99	2.54	1.12
smb_t	.070	.186	-.010	-.139	-.070	-1.342
t	1.99	1.98	-0.37	-3.23	-0.78	-1.09
r_{t-1}^{10}	-.037	-.049	-.036	-.062	-.058	-.220
t	-2.09	-2.54	-10.89	-13.44	-5.05	-2.08
lev_{t-1}^i	1.385	-.428	.208	.527	.651	.376
t	13.12	-0.68	3.18	7.38	3.72	0.16
VIX_{t-1}	.001	.000	.004	.009	.031	.132
t	0.27	0.09	4.12	4.22	4.01	3.07
defprem_{t-1}	-.365	-.473	-.306	-.391	-.392	-.314
t	-2.56	-4.11	-18.47	-15.80	-6.58	-0.47
r_{t-1}^{SP}	-.005	-.005	-.005	-.004	-.008	.013
t	-4.05	-4.25	-9.36	-5.18	-2.76	0.48
adjusted R^2	0.371	0.411	0.335	0.323	0.263	0.294
N	4	56	275	245	90	18

Table 9: Relation Between Changes in Credit Spreads and Interest Rate Levels

For each industrial bond i having at least 25 monthly quotes CS_t^i over the period July 1988 to December 1997, we estimate the following regression: $\Delta CS_t^i = \alpha + \beta_1^i r_{t-1}^{10} + \epsilon_t^i$. Quotes are discarded whenever a bond has less than 4 years to maturity. Average OLS parameter estimates are reported in Panel I. Panel II shows averages for a short maturity subsample where quotes are discarded whenever a bond has more than 9 years to maturity. Panel III shows averages for a long maturity subsample where quotes are discarded whenever a bond has less than 12 years to maturity. Associated t-statistics for each average appear immediately beneath.

	Leverage Groups					
	<15%	15-25%	25-35%	35-45%	45-55%	>55%
I. Whole Sample						
intercept	-.038	-.044	-.086	-.095	-.114	-.285
t	-2.56	-3.57	-4.96	-3.67	-4.01	-2.57
r_{t-1}^{10}	.006	.006	.011	.012	.015	.040
	2.51	3.50	4.74	3.33	3.53	2.62
adjusted R^2	-.016	-.012	-.010	-.008	-.008	-.008
N	100	162	138	123	91	74
II. Short Maturity Subsample						
intercept	-.093	-.102	-.153	-.146	-.098	-.413
	-3.96	-5.57	-4.88	-3.28	-2.06	-2.34
r_{t-1}^{10}	.013	.014	.020	.018	.010	.058
	3.67	5.51	4.61	2.96	1.47	2.38
adjusted R^2	-.014	-.015	-.008	-.009	-.014	-.010
N	53	91	65	64	47	46
III. Long Maturity Subsample						
intercept	.002	.011	-.028	-.081	-.104	-.088
	0.11	0.46	-1.02	-1.30	-3.42	-2.88
r_{t-1}^{10}	.000	-.002	.003	.009	.015	.012
	-0.10	-0.51	0.89	1.19	3.34	2.71
adjusted R^2	-.014	-.012	-.010	-.003	-.008	-.011
N	33	54	50	45	33	27

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